

10.3 The challenges of food supply

Supplying food to an ever-growing global population presents a number of challenges that we have to overcome to ensure food security. Both human and natural factors impact food supply and include water security, land degradation, natural disasters, pests, and diseases which can all reduce the amount of food being produced in a region.

Table 10.1: Human and natural factors affecting food supply

Factor	Human and/or natural factors	Description of factors affecting food supply
Water shortages	Human and natural factors	A lack of water due to the local climate or drought can result in crop failure and the inability to raise livestock. This can be a natural factor where lack of rainfall reduces water availability. The overuse of water by humans can also cause water shortages.
Land degradation and desertification		Poor farming practices leads to desertification, land degradation, soil exhaustion, and soil erosion. This reduces the ability of the land to support crops and food for raising livestock, limiting food production. However, climate change and reduced rainfall can also naturally cause areas to become more arid which also results in land degradation.
Agricultural pests and diseases		Pests such as locusts or rodents can consume large amounts of food from the fields each day. Fungal diseases can destroy crops and food being stored. These natural factors are made worse due to monocropping which supports the spread of pests.
Climate change		As the climate shifts, the change in temperature and the amount of rain (including extreme weather events such as storms, floods and droughts) impacts the crops that grow in certain climate conditions. This results in the decline in food production in previously successful agricultural regions. Climate change is both a natural phenomenon and one that is enhanced due to human activities like burning fossil fuels.
Population growth	Human factors	Ongoing population growth means a higher food demand. Where rapid population growth and subsistence farming occur, typically in LICs, the challenge of feeding people is even greater.
Demand for biofuel		Biofuel is an energy source made from plant and animal materials. As human demand for this renewable energy source increases, fewer crops for food and farmland are available.
Poverty		When people have too little money, they are not able to purchase food. Without food, they are unable to work, and therefore struggle to make money. This poverty cycle is hard to break and may result in the population being dependent on food aid.
Transport and storage		A lack of storage for food products during periods of excess production means that food cannot be stored for use during times of shortage. In addition, the lack of ability to transport food to areas where it is needed prevents food from being supplied. This could be due to the damage of infrastructure after a natural disaster or period of conflict, or due to low levels of infrastructure development in a country with limited economic resources.
War and political unrest		Where war and political unrest occur, farmers are often driven off their lands. This results in a reduction in food production in a region, and a difficulty in supplying food.



Source A: Cartoon showing a farmer putting his rain gauge up for sale

TASK 1: Study Source A

- a** In pairs, discuss the message in the cartoon. Then, create a flow chart to show how the lack of need for a rain gauge on the farm impacts the farmer and food production in a region.
- b** Research food production in your region. Have recent weather events, or changes to local weather patterns affected local food production? In pairs, write a brief summary of what you find and share the information with the class.

The problems caused by food insecurity

Food insecurity has significant impacts on the population, economy, and environment. There are four main impacts of food insecurity:

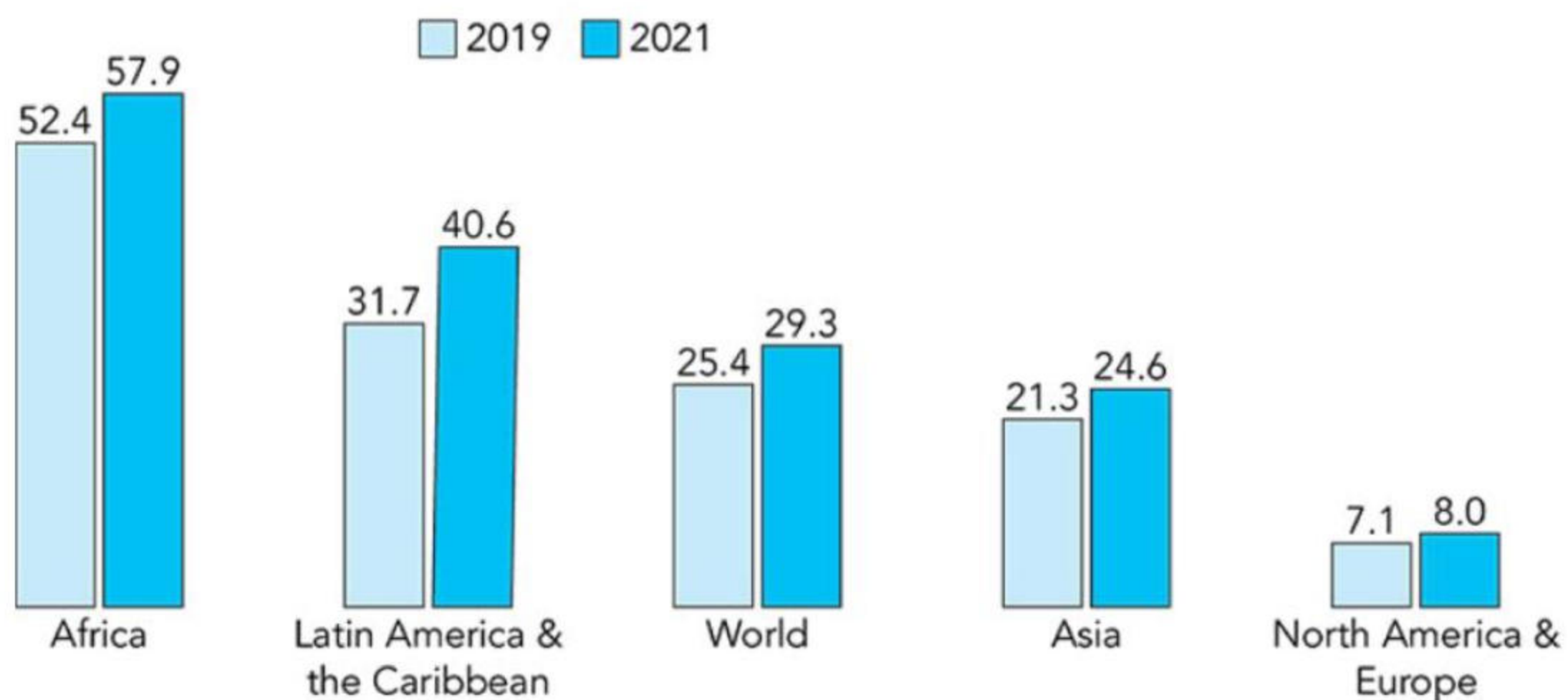
- **An increase in the number of people living in poverty:** adults living in food-insecure households are more likely to suffer from poor health. This means that they are unable to work and earn an income. This results in them living in poverty.
- **Slowing down or shrinking of the economy resulting in increased food prices:** with the workforce suffering from food shortages and unable to work, the economy slows as the workforce fails. This results in the prices of items increasing as they become scarcer.
- **Political destabilisation with food-related unrest:** hunger can drive a population to lead protests against food scarcity and costs. This causes political unrest within a region or country.
- **Population decline due to out-migration or death:** as food becomes scarcer, people will move away in search of a more secure food source. If they are unable to move, then they are likely to starve due to the lack of food.

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Food insecurity can cause **malnutrition** and this can lead to starvation. When suffering from malnutrition, a person is less resistant to disease and is more likely to fall ill. They are then unable to work which further decreases their ability to earn money to buy food or grow the food themselves.

The impacts of food insecurity are most intensely felt in LICs, where lower levels of economic development result in less money available to purchase food. However, food insecurity does occur in MICs and HICs too. This can be triggered by economic slowdown, a natural disaster destroying crops and infrastructure, or a global outbreak of disease which can impact the transport and delivery of food between regions or countries. In addition, in MICs and HICs, there are areas of poverty within the country. These areas can struggle with food security due to the shortage of money to purchase food.

Share of population experiencing moderate/severe food insecurity, by region (in percent)



Source **B**: The impacts of COVID-19 on world hunger

► Image Description

TASK 2: Study Source **B**

The COVID-19 pandemic started in 2019, and by 2021, at least 828 million people were affected by hunger. This was 150 million more than 2019 and 46 million more than 2020.

- Which region** experienced the largest change in food insecurity between 2019 and 2021?
- Describe the change in the number of people suffering from hunger from 2019 to 2021.**
- In pairs, discuss the change that you observe in food insecurity by region, including the following areas:
 - Did any of the data surprise you?
 - Which regions had lower increases in food security than the world average, and which areas had higher increases?
 - Give three suggestions for why these trends occurred.

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Managing food insecurity

There are different strategies and techniques for managing food insecurity. There are short-term strategies such as **food aid** and long-term strategies such as training farmers how to protect soils and farmlands.

Food aid

Food aid is when a country receives food from another country during periods of food shortages. This can be as a donation or under fair terms to help the country in need. In order to help people in need, non-governmental organisations (NGOs) like the World Food Programme (WFP) have also been established to provide food to food-insecure regions; for example, the WFP is helping move food into South Sudan which is struggling with both long-term drought and civil conflict.

Food aid should not be seen as a long-term solution; it should be implemented in response to an event that has threatened food in the short term. Food aid is successful when it is able to reach the people in need and when it is not used for too long. However, some regions that require food aid may be at war which means that food does not reach the intended population. Furthermore, if the area is suffering from a drought which lasts for a long time, then food aid has to continue and this can develop a population that becomes dependent on the food aid and are no longer able to provide for themselves.

Improving soil protection

Long-term solutions look at how to establish farming practices to ensure that people are able to feed themselves in the future. This is done through learning how to protect the soils and to prevent the soils from becoming depleted, and therefore unable to support crop growth. Table 10.2 outlines some key techniques.

Table 10.2: Techniques for protecting farmlands from degradation and soil erosion

Techniques used to improve soil protection during farming	
Protecting soils from desertification	

	• Water management, including minimising overuse of water to protect water resources. • Selecting crops that have low water requirements. • Preventing total loss of vegetation, which protects soils from soil erosion, through preventing overgrazing or harvesting of vegetation.
Training farmers	Training farmers in the following areas: • How to correctly irrigate to protect water resources and soil quality. • How to rotate crops to allow soil to rest between harvests and to vary the crops being grown in a field. This helps to protect soil nutrients and soil fertility. • How to prevent overgrazing by not overstocking an area. • How to protect groundwater resources from over-abstraction (where the amount of water taken from the ground is greater than the amount of water falling as rain), which further dries out surface areas. • How to select the correct crops for the climate and use organic fertilisers to help plants grow. • How to protect the natural vegetation which protects soils from wind and water erosion. This can be done by planting up areas that have had their vegetation removed or protecting areas that still have their natural vegetation in place. • The best techniques to protect soils from soil erosion such as planting wind breaks, terracing slopes, contour ploughing, and managing rainwater runoff (Figure 10.1).

Figure 10.6: Different soil erosion management techniques



Steep-like terraces built on slopes prevents water runoff carrying away soil



Crops are planted to follow the contour of the land, which slows runoff and reduces erosion



Rows of trees planted between fields slows the wind and reduces wind erosion

TIP

When considering protecting soils from degradation, remember that not all soils are the same. Some soils are naturally less fertile and less likely to support plant growth than others. In addition, soils require nutrients, water, and sufficient sunlight to support good plant growth. This means that there are regions of the world that are natural deserts which are not caused by poor farming practices.

The success of strategies and techniques in protecting soils from degradation and erosion depends on a wide range of factors, including the environment, climate, level of poverty, and willingness of the population to learn and change their practices and even their culture. For example, where large livestock herds represent family wealth, it is difficult to change the behaviour of the people. To protect the soils from overgrazing, smaller herds are required, however, the people do not wish to give up their wealth.



Source : Sheep-grazing plants in a dry region with limited vegetation growth

- a** Define the term desertification.
- b** Does Source C show potential desertification? Support your answer with observations from the image and information in Table 10.2.
- c** Explain what a farmer could do to protect the soils in a dry region like this.

THINK LIKE A GEOGRAPHER

Geographers need to think of climate when considering food supply. They need to think about the soils, the seasons, rainfall, and the length of day to consider the production of food. In addition, climate change is adding to the challenge of understanding food security.

How would an increase or decrease in rainfall impact crop growth? As a geographer, how would you identify where in the world there might be an increase or decrease in rainfall over time? How long would it take for the impact to be felt?